

Answers

1. Given $\vec{v}_1 = (1, 0)$ and $\vec{v}_2 = (0, 1)$, what is $\vec{v}_1 \cdot \vec{v}_2$

Answer: $1 * 0 + 0 * 1 = 0$

2. Given $\vec{v}_1 = (1, 2)$ and $\vec{v}_2 = (2, 1)$, what is $\vec{v}_1 \cdot \vec{v}_2$

Answer: $1 * 2 + 2 * 1 = 4$

3. Given $\vec{v}_1 = (1, 2)$ and $\vec{v}_2 = (3, 4)$, what is $\vec{v}_1 \cdot \vec{v}_2$

Answer: $1 * 3 + 2 * 4 = 11$

4. Given $\vec{v}_1 = (2, 3)$ and $\vec{v}_2 = (-3, 2)$, what is $\vec{v}_1 \cdot \vec{v}_2$

Answer: $2 * -3 + 3 * 2 = 0$

5. Given $\vec{v}_1 = (2, 3, 1)$ and $\vec{v}_2 = (1, 1, 1)$, what is $\vec{v}_1 \cdot \vec{v}_2$

Answer: $2 * 1 + 3 * 1 + 1 * 1 = 6$

6. The dot product is also known by what names?

Answer: *Scalar projection, scalar product.*

7. Is the dot product associative?

Answer: *No, for the dot product to be associative, $A \cdot (B \cdot C) = (A \cdot B) \cdot C$ must be true, but the dot product between 2 vectors produces a scalar, and you can't take the dot product of a vector and a scalar.*

8. Commutative? Show using any 2 vectors.

Answer: i. $(1, 2) \cdot (3, 4) = 1 * 3 + 2 * 4 = 11$
ii. $(3, 4) \cdot (1, 2) = 3 * 1 + 4 * 2 = 11$

9. If $\vec{v}_1 \cdot \vec{v}_2 = 0$, what does this signify geometrically?

Answer: *They are orthogonal to each other.*

10. If $\hat{v}_1 \cdot \hat{v}_2 < 0$, what does this signify geometrically?

Answer: *They roughly point in opposite directions*

11. As $\hat{v}_1 \cdot \hat{v}_2$ approaches 1, what does this signify geometrically?

Answer: As the dot product approaches one, the vectors become more and more aligned until they become the exact same vector.

12. Given $\hat{v}_1$, and $\hat{v}_2$, what is the possible range of values for $\hat{v}_1 \cdot \hat{v}_2$? Why?

Answer: The range of possible values is from -1 to 1. Recall the formula for the dot product. So given 2 vectors, $\vec{v}_1 \cdot \vec{v}_2 = |\vec{v}_1| |\vec{v}_2| \cos \theta$. Since the magnitude of $\hat{v}_1$ and $\hat{v}_2$ is, by definition 1, since these are unit vectors, then $\hat{v}_1 \cdot \hat{v}_2 = \cos \theta$, and the cosine function returns values in the range -1 to 1.

13. How is the dot product related to the angle between 2 unit vectors?

Answer: The dot product of 2 unit vectors is related to the cosine of the angle between them.

14. What is the relationship between $\vec{v}_1 \cdot \vec{v}_1$ and $|\vec{v}_1|^2$?

Answer: The dot product of a vector with itself is equal to the square of the magnitude.

15. Given 2 vectors, $\vec{v}_1$ and $\vec{v}_2$, how would you determine if they're orthogonal?

Answer: Take the dot product, if it's 0, they're orthogonal.

16. How would you use the dot product to implement a "lock-on" targeting system?

Answer: So if you're building a game and wanted to implement some sort of autolock or "lock on" system, whereby your targeting cursor immediately latches onto nearby targets, you could do something like this. You'd have your aiming direction, let's call that $\hat{a}$, and the vectors from the player to each enemy, call those vectors $\hat{e}_i$, then this could be done by calculating the dot product between $\hat{a}$ and $\hat{e}_i$, filtering out anything that didn't meet some predetermined minimum threshold, and then simply choosing the highest result. This would ensure that you've chosen the target nearest to the current targeting direction, and your game logic could adjust the target direction to exactly match.

17. Let's imagine you're building a tower defense game with an automated tower, firing at a group of enemies. Explain how you might use the dot product to choose which targets to fire at.

Answer: This is quite similar to the previous example. In this case, your automated tower already has a direction, let's call it $\hat{a}$, and in essence, you're trying to shoot at whatever requires the least amount of adjustment to the tower's direction. So you'd

calculate the direction to each enemy, call that $\hat{e}_i$, calculate the dot product between $\hat{a}$ and $\hat{e}_i$. Once that's done, you choose the highest value, so that the tower can prioritize firing at that specific enemy.